

Walking Difficulty and Depression Severity in U.S. Adults: Is the Association Modified by Anxiety?

GPH-GU 2450 Intermediate Epidemiology Final Project Report

Boyuan Zhang

Dataset: NHANES 2021-2023

Abstract

Background: Depression is a major public health issue, and physical functioning limitations may be related to worse mental health. This study examined whether difficulty walking or climbing steps was associated with depression severity among U.S. adults and whether this association varied by anxiety frequency.

Methods: This cross-sectional analysis used NHANES 2021-2023 data among U.S. adults aged 20 years and older. Depression was measured using the PHQ-9 as a continuous score and as a binary outcome defined as PHQ-9 score ≥ 10 . The main exposure was self-reported difficulty walking or climbing steps. Univariable and bivariable analyses described the sample and crude associations. Multivariable logistic and linear regression models estimated adjusted associations with and without a walking difficulty by anxiety frequency interaction term.

Results: The analytic population included 7,809 adults. Among participants with complete PHQ-9 data, 12.7% met criteria for depression. Walking difficulty was associated with higher depression prevalence and higher mean PHQ-9 score. Adjusted logistic and linear regression models showed a graded pattern, with more severe walking difficulty associated with higher odds of depression and higher mean PHQ-9 score. Anxiety frequency was strongly associated with both depression outcomes. Interaction estimates suggested possible variation by anxiety frequency, but some estimates were imprecise.

Conclusions: Greater walking difficulty was independently associated with higher depression severity in U.S. adults. Findings support considering physical functioning and anxiety symptoms together when identifying adults at higher risk of depression.

Introduction

Depression is an important public health problem because it is common, disabling, and unequally distributed across the U.S. population. In 2020, approximately 18.5% of U.S. adults reported that a health care provider had diagnosed them with depression (Lee et al., 2023). Prior research also shows that depression differs by sex, age, race/ethnicity, education, financial assets, and geography, with higher rates among women, younger adults, and adults with fewer socioeconomic resources (Ettman et al., 2024; Gonzalez et al., 2010; Mojtabai et al., 2016).

Physical functioning limitations are also important to study in relation to depression. Prior studies suggest that lower physical activity, greater sedentary behavior, walking impairment, and disability domains are associated with higher depression risk or more depressive symptoms (Becofsky et al., 2013; Meng et al.,

2024; Meshkat et al., 2025; Rezvani et al., 2022; Thompson et al., 2012; Zhang & Jiang, 2023). For example, NHANES-based research found that physical activity and sedentary behavior were associated with depression among U.S. adults, and another NHANES study found that depressive symptoms were strongly associated with multiple disability domains, including walking difficulty (Meng et al., 2024; Meshkat et al., 2025).

Studies from other populations also support this pattern, showing that better walking ability is associated with lower odds of depressive symptoms and that walking impairment is related to greater mental health burden (Rezvani et al., 2022; Zhang & Jiang, 2023).

Anxiety symptoms are also relevant because they frequently co-occur with depression and may worsen the mental health consequences of physical limitations. Prior work examining anxiety, depression, and disability suggests that anxiety and depressive symptoms are both associated with disability, and their combination may be especially important (Brenes et al., 2008). However, it remains unclear whether the relationship between difficulty walking or climbing steps and depression differs according to how often an individual feels worried, nervous, or anxious. This study addresses that gap using recent NHANES 2011-2018 data.

Research Question and Hypothesis

Research question: Among U.S. adults aged 20 years and older, is greater difficulty walking or climbing steps independently associated with higher severity of depression, and does the strength of this association vary based on how frequently an individual feels worried, nervous, or anxious, after adjusting for smoking status, diabetes condition, and high cholesterol level?

Hypothesis: Greater difficulty walking or climbing steps is independently associated with increased depression severity after adjustment for health-related covariates. Specifically, more severe walking difficulty is expected to correspond to higher PHQ-9 scores and greater likelihood of binary depression. In addition, the association between walking difficulty and depression is expected to be stronger among individuals who frequently feel worried, nervous, or anxious than among those who report anxiety symptoms less frequently.

Methods

Study Design

This study used data from the National Health and Nutrition Examination Survey (NHANES), a program of the National Center for Health Statistics within the Centers for Disease Control and Prevention. NHANES assesses the health and nutritional status of the U.S. population using interviewer-administered questionnaires, physical examinations, and laboratory testing. The survey uses a complex, multistage, stratified, clustered probability sampling design to produce estimates representative of the civilian, non-institutionalized U.S. population (Centers for Disease Control and Prevention, 2026). Data from the August 2021 through August 2023 wave were used for this analysis.

Study Population

The NHANES 2021-2023 wave included 15,560 participants who completed the household interview and 14,300 participants who completed the medical examination component. The present analysis used a subsample of 8,153 participants and was restricted to adults aged 20 years and older ($n = 7,809$), because several covariates such as education level and marital status were assessed only among adults in this age range. Although the PHQ-9 depression module was administered to adults aged 18 years and older, restricting the study population to age ≥ 20 years ensured consistency across analytic variables. Responses coded as refused or do not know were treated as missing. Descriptive and bivariable analyses used available non-missing data for each variable; multivariable models used complete cases for the variables included in each model (Centers for Disease Control and Prevention, 2026).

Main Outcome

The primary outcome was depression severity measured with the Patient Health Questionnaire-9 (PHQ-9). The PHQ-9 includes nine items assessing depressive symptoms experienced during the prior two weeks. Each item is scored from 0 to 3, and item scores are summed to create a total score ranging from 0 to 27, with higher scores indicating greater depression severity. The PHQ-9 was analyzed both as a continuous score and as a binary outcome, with scores ≥ 10 classified as probable depression. The tenth NHANES item assessing difficulty caused by these problems was not included in the PHQ-9 score calculation (Kroenke et al., 2001).

Main Exposure

The main exposure was difficulty walking or climbing steps, assessed with a physical functioning questionnaire item asking participants whether they had difficulty walking or climbing steps. Response options included no difficulty, some difficulty, a lot of difficulty, and cannot do this at all. This variable was maintained as a four-category ordinal exposure in the multivariable models, with no difficulty used as the reference category. For bivariable summaries, difficulty was also presented as no difficulty versus any difficulty.

Covariates and Effect Measure Modifier

Sociodemographic covariates included gender, age group, race/ethnicity, marital status, education level, country of birth, and family income-to-poverty ratio. Health-related covariates included high cholesterol and diabetes. Smoking status was described in the univariable analysis; however, because the denominator for smoking status was restricted to participants who had smoked at least 100 cigarettes in their lifetime, it was not retained in the final multivariable tables shown in this report. The proposed effect measure modifier was frequency of feeling worried, nervous, or anxious, dichotomized as frequent (daily or weekly) versus infrequent (monthly, a few times a year, or never). The family income-to-poverty ratio was interpreted using Census Bureau poverty guidance (U.S. Census Bureau, 2026).

Statistical Analysis

Univariable analyses were used to describe the analytic population. Continuous variables were summarized using means, standard deviations, ranges, medians, and interquartile ranges as appropriate. Categorical variables were summarized using frequencies and percentages. Bivariable analyses examined associations between walking difficulty, depression outcomes, and covariates. Chi-square tests or Fisher exact tests were used for categorical variables, independent-samples t-tests were used for continuous outcomes with binary predictors, ANOVA was used for categorical predictors with more than two levels, and correlation tests were used for continuous age and PHQ-9 score.

Multivariable regression analyses were conducted for both depression outcomes. Logistic regression estimated odds ratios (ORs) and 95% confidence intervals (CIs) for binary depression. Linear regression estimated beta coefficients and standard errors for continuous PHQ-9 score. Full model 1 included

walking difficulty and covariates. Full model 2 included the same variables plus a walking difficulty by anxiety frequency interaction term to assess effect measure modification. Statistical significance was evaluated using $\alpha = 0.05$.

Results

Univariable Analysis

The analytic sample included 7,809 U.S. adults aged 20 years and older from NHANES 2021-2023. PHQ-9 scores were missing for 2,599 participants (33.3%); therefore, denominators vary across characteristics presented in Table 1. Among those with complete data, 55.3% were female. Mean age was 53.6 years, and median age was 57 years. The sample was predominantly non-Hispanic White (58.3%). Most participants had at least some college education and were married or living with a partner. For the main exposure, 66.3% reported no difficulty walking or climbing steps, while 33.7% reported some level of difficulty. The mean PHQ-9 score was 4.0, and 12.7% met criteria for depression. The distribution of PHQ-9 score was right-skewed, and observed outliers were retained because they were within the valid PHQ-9 range of 0-27.

Table 1. Univariable Analysis of U.S. Adults Aged 20 Years and Older, National Health and Nutrition Examination Survey (NHANES), United States, 2021-2023 (n = 7,809)

	n (%)
Sociodemographic characteristics	
Gender (N = 7809)	
Male	3488 (44.7%)
Female	4321 (55.3%)
Age (years) (N = 7809)	
Mean (SD, range)	53.6 (17.5, 20 - 80)
Median age (IQR)	57 (38 - 68)
Race/Ethnicity (N = 7809)	
Mexican American	545 (7.0%)
Other Hispanic	780 (10.0%)
Non-Hispanic White	4555 (58.3%)
Non-Hispanic Black	995 (12.7%)
Non-Hispanic Asian	422 (5.4%)
Other/Multi-Racial	512 (6.6%)
Education level (N = 7783)	
Less than High School	1039 (13.3%)
High School Graduate/GED	1749 (22.5%)
Some College or College Graduate	4995 (64.2%)
Marital Status (N = 7783)	
Married/Living with partner	4136 (53.1%)
Widowed/Divorced/Separated	2022 (26.0%)
Never Married	1625 (20.9%)
Country of Birth (N = 7793)	
Born in 50 US states or Washington DC	6219 (79.8%)
Others	1574 (20.2%)
Ratio of family income to poverty (N = 6489)	
<1	1037 (16.0%)
1	7 (0.1%)
>=1	5445 (83.9%)
Depression (PHQ-9) Score (continuous) (N = 5210)	
Mean (SD, range)	4.0 (4.7, 0 - 26)
Median age (IQR)	2 (1 - 6)
Depression (N = 5210)	
No	4549 (87.3%)
Yes	661 (12.7%)
Difficulty Walking or Climbing Steps (N = 7797)	
No Difficulty	5167 (66.3%)
Some Difficulty	1848 (23.7%)
A Lot of Difficulty	683 (8.7%)
Cannot Do at All	99 (1.3%)
High Cholesterol (N = 7755)	
Yes	3052 (39.4%)
No	4703 (60.6%)

Smoking Status (N = 3233)	
Every Day	948 (29.3%)
Some Days	235 (7.3%)
Not at All	2050 (63.4%)
Diabetes Condition (N = 7806)	
Yes	1070 (13.7%)
No	6466 (82.8%)
Borderline	270 (3.5%)
Frequency of Feeling Worried/Nervous/Anxious (N = 7785)	
Infrequent	4656 (59.8%)
Frequent	3129 (40.2%)

SD = standard deviation; IQR = interquartile range

Note. Numbers may not sum to the total sample size because of missing data. Education and marital status were assessed only among adults aged 20 years and older. Smoking status was assessed only among participants who reported ever smoking 100 or more cigarettes in their lifetime; therefore, the denominator for this variable is n = 3,233. PHQ-9 scores were missing for n = 2,599 (33.3%) participants. Income to poverty ratio was top-coded at 5.0 (values ≥ 5.0 coded as 5). Depression defined as PHQ-9 score ≥ 10 . Frequency of feeling worried, nervous, or anxious was dichotomized as "infrequent" (monthly, a few times a year, never) and "frequent" (daily, weekly).

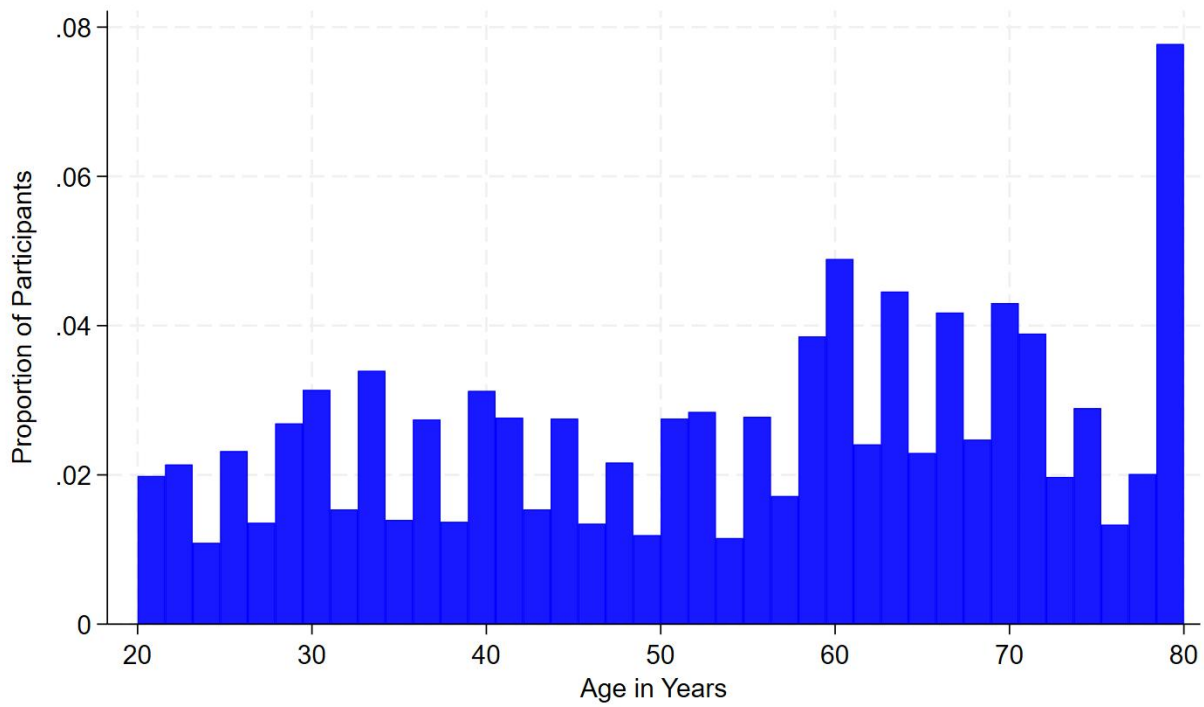


Figure 1. Histogram of age distribution among U.S. adult participants aged 20 years and older in the NHANES 2021-2023 survey (n = 7809).

Note. NHANES 2021-2023 = National Health and Nutrition Examination Survey, August 2021-August 2023. Age was top-coded at 80 years; participants aged 80 years and older are represented as age 80.

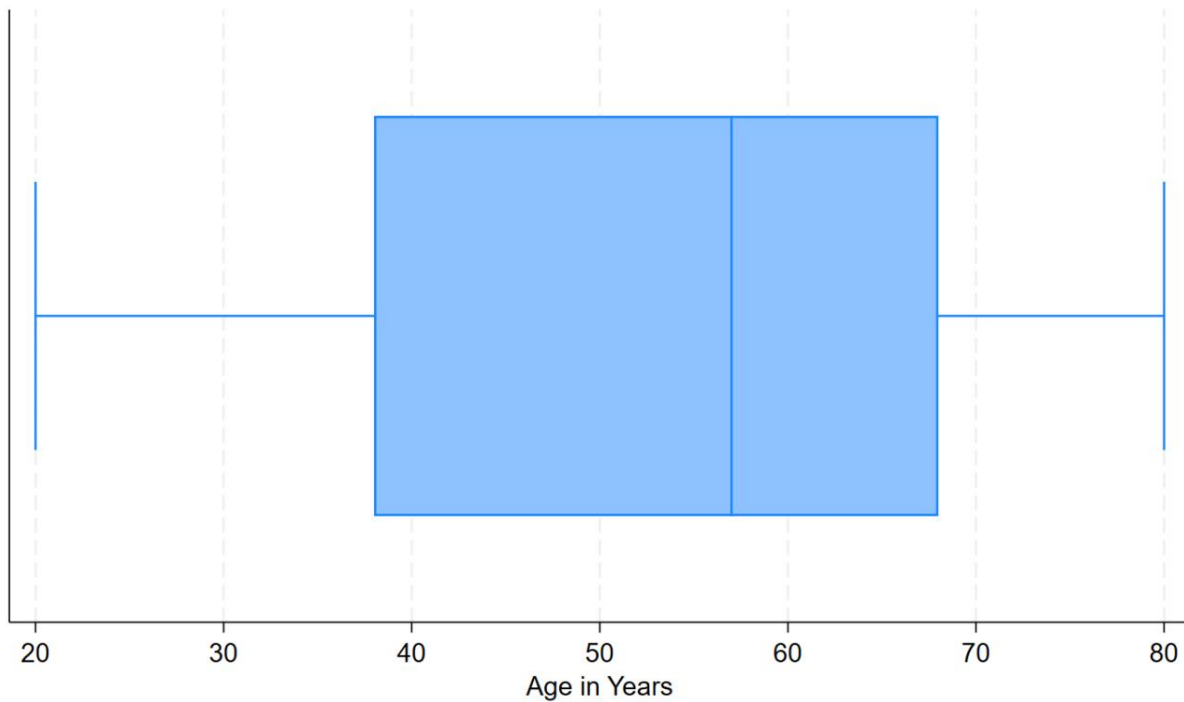


Figure 2. Boxplot of age distribution among U.S. adult participants aged 20 years and older in the NHANES 2021-2023 survey (n = 7809).

Note. NHANES 2021-2023 = National Health and Nutrition Examination Survey, August 2021-August 2023. Age was top-coded at 80 years; participants aged 80 years and older are represented as age 80.

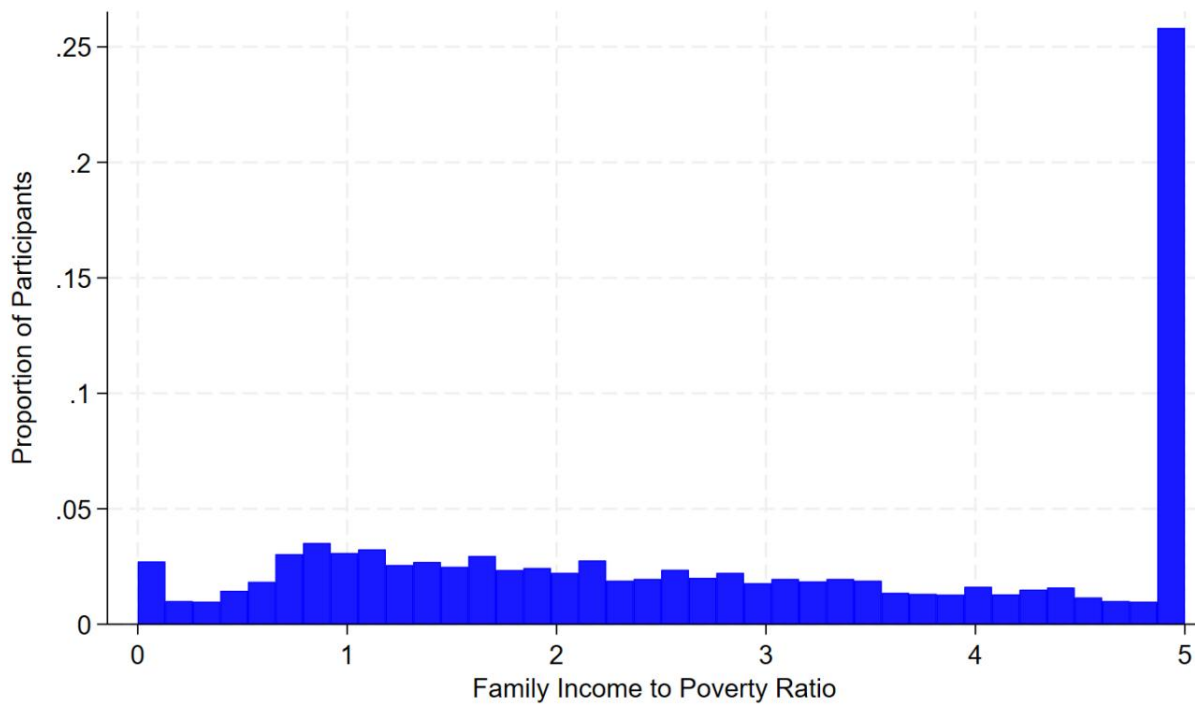


Figure 3. Histogram of family income to poverty ratio distribution among U.S. adult participants aged 20 years and older in the NHANES 2021-2023 survey (n = 6489).

Note. NHANES 2021-2023 = National Health and Nutrition Examination Survey August 2021-August 2023. Values ≥ 5.0 are top-coded as 5.

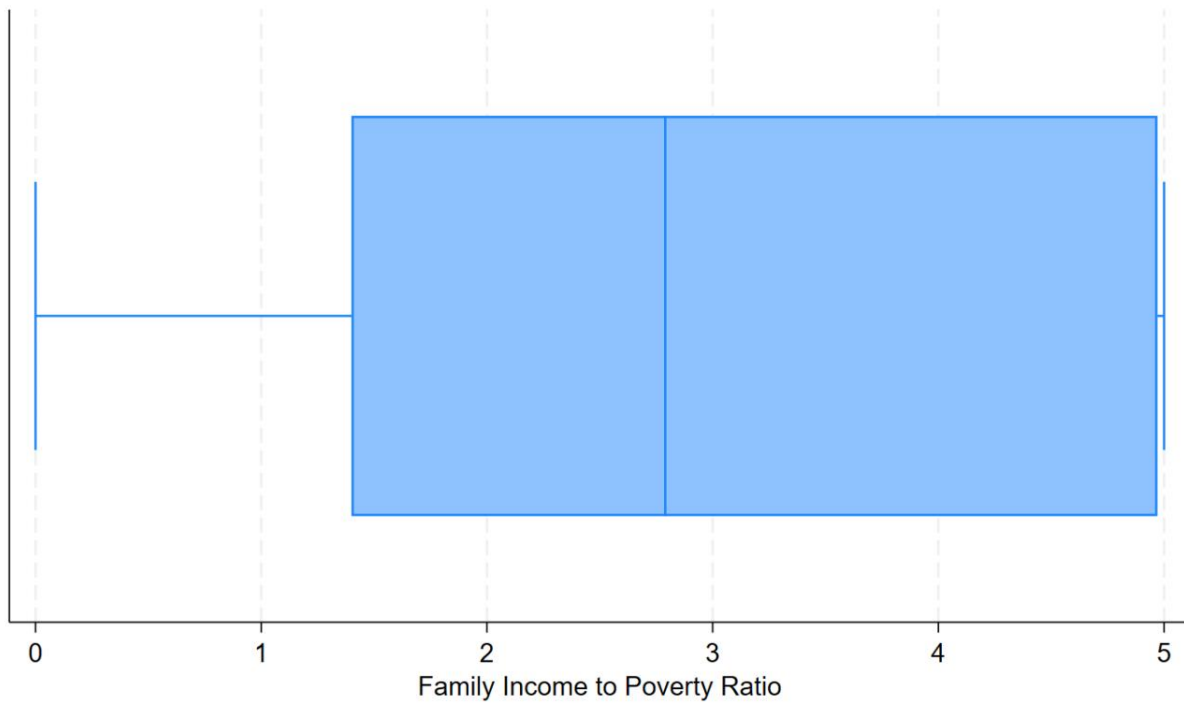


Figure 4. Boxplot of family income to poverty ratio distribution among U.S. adult participants aged 20 years and older in the NHANES 2021-2023 survey (n = 6489).

Note. NHANES 2021-2023 = National Health and Nutrition Examination Survey August 2021-August 2023. Values ≥ 5.0 are top-coded as 5.

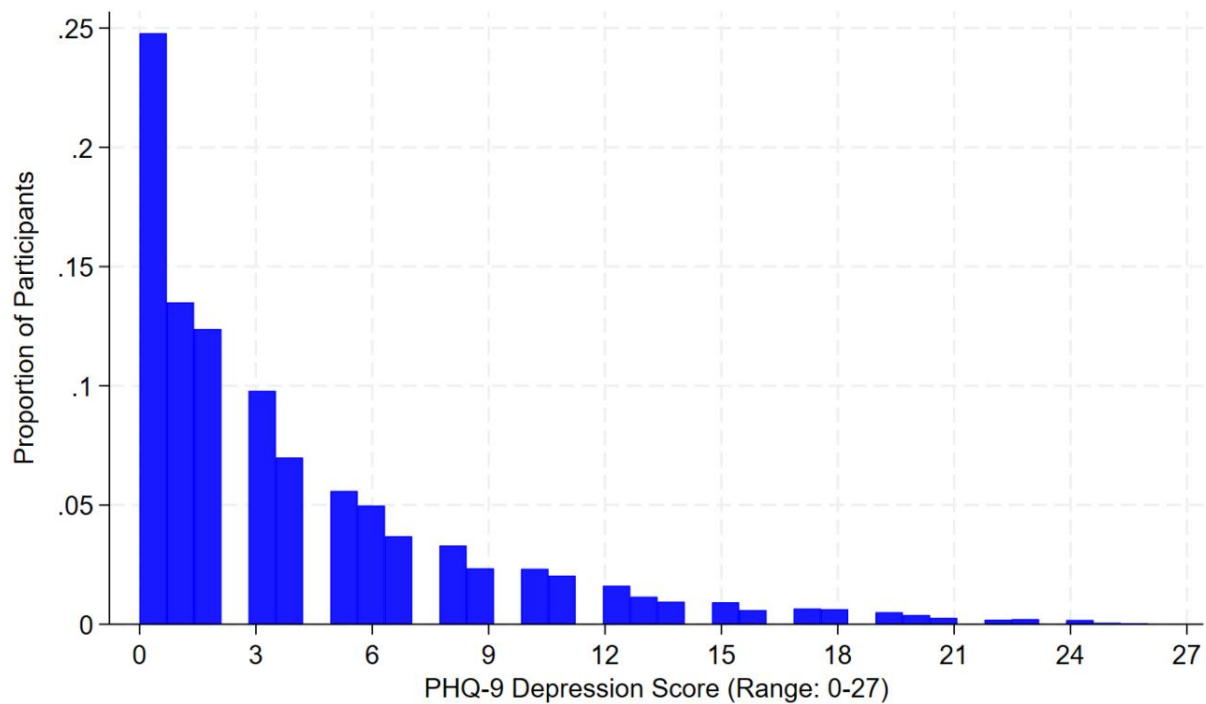


Figure 5. Histogram of PHQ-9 depression score distribution among U.S. adult participants aged 20 years and older in the NHANES 2021-2023 survey (n = 5210).

Note. NHANES 2021-2023 = National Health and Nutrition Examination Survey August 2021-August 2023. PHQ-9 = Patient Health Questionnaire-9. PHQ-9 scores were calculated for participants with complete responses to all nine items. Scores range from 0 to 27, with higher scores indicating greater depression severity.

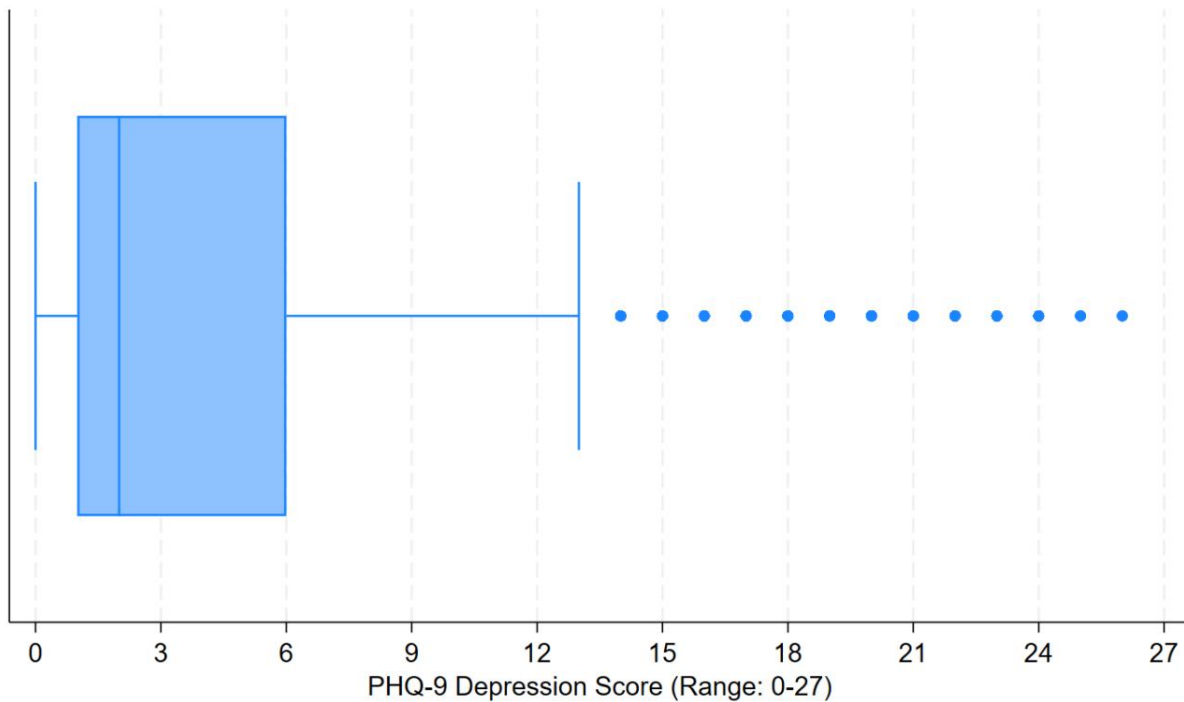


Figure 6. Boxplot of PHQ-9 depression score distribution among U.S. adult participants aged 20 years and older in the NHANES 2021-2023 survey (n = 5210).

Note. NHANES 2021-2023 = National Health and Nutrition Examination Survey August 2021-August 2023. PHQ-9 = Patient Health Questionnaire-9. PHQ-9 scores were calculated for participants with complete responses to all nine items. Scores range from 0 to 27, with higher scores indicating greater depression severity. Outliers beyond the upper whisker were verified against the original records and retained because they represented plausible scores.

Bivariable Analysis

Table 2. Bivariable associations between difficulty walking or climbing steps and PHQ-9 depression score and key sociodemographic characteristics and covariates, NHANES 2021-2023 (n = 7,809)

		Difficulty walking or climbing steps		p-value
Sociodemographic Characteristics	Total n (%)	No n (%)	Yes n (%)	
Gender (N = 7809)				
Male	3488 (44.7%)	2473 (71.0%)	1008 (29.0%)	<0.001 ²
Female	4321 (55.3%)	2694 (62.4%)	1622 (37.6%)	
Age (years) (N = 7809)				
Mean (SD, range)	53.6 (17.5, 20 - 80)	49.1 (17.2, 20 - 80)	62.5 (14.4, 21 - 80)	<0.001 ³
Median age (IQR)	57 (38 - 68)	48 (34 - 64)	65 (55 - 74)	
Race/Ethnicity (N = 7809)				
Mexican American	545 (7.0%)	418 (76.7%)	127 (23.3%)	<0.001 ²
Other Hispanic	780 (10.0%)	533 (68.7%)	243 (31.3%)	
Non-Hispanic White	4555 (58.3%)	2917 (64.1%)	1632 (35.9%)	
Non-Hispanic Black	995 (12.7%)	650 (65.3%)	345 (34.7%)	
Non-Hispanic Asian	422 (5.4%)	339 (80.7%)	81 (19.3%)	
Other/Multi-Racial	512 (6.6%)	310 (60.6%)	202 (39.4%)	
Education level (N = 7783)				
Less than High School	1039 (13.3%)	549 (52.9%)	489 (47.1%)	<0.001 ²
High School Graduate/GED	1749 (22.5%)	1070 (61.2%)	679 (38.8%)	
Some College or College Graduate	4995 (64.2%)	3535 (70.8%)	1457 (29.2%)	
Marital Status (N = 7783)				
Married/Living with partner	4136 (53.1%)	2919 (70.6%)	1214 (29.4%)	<0.001 ²
Widowed/Divorced/Separated	2022 (26.0%)	1005 (49.7%)	1016 (50.3%)	
Never Married	1625 (20.9%)	1227 (75.5%)	398 (24.5%)	
Country of Birth (N = 7793)				
Born in 50 US states or Washington DC	6219 (79.8%)	4022 (64.7%)	2194 (35.3%)	<0.001 ²
Others	1574 (20.2%)	1137 (73.3%)	436 (27.7%)	
Ratio of family income to poverty (N = 6489)				
<=1				<0.001 ²
>1	1044 (16.1%) 5445 (83.9%)	573 (54.9%) 3731 (68.5%)	470 (45.1%) 1713 (31.5%)	
Depression (PHQ-9) Score (continuous) (N = 5210)				
Mean (SD, range)	4.0 (4.7, 0 - 26)	3.3 (4.2, 0 - 26)	5.5 (5.3, 0 - 26)	<0.001 ³
Median age (IQR)	2 (1 - 6)	2 (0 - 5)	4 (1 - 8)	
Depression (N = 5210)				
No	4549 (87.3%)	3175 (69.8%)	1372 (30.2%)	<0.001 ²
Yes	661 (12.7%)	312 (47.2%)	349 (52.8%)	
High Cholesterol (N = 7755)				
Yes	3052 (39.4%)	1663 (54.6%)	1385 (45.4%)	<0.001 ²

		Difficulty walking or climbing steps		p-value
Sociodemographic Characteristics	Total n (%)	No n (%)	Yes n (%)	
No	4703 (60.6%)	3482 (74.1%)	1215 (25.9%)	
Diabetes Condition (N = 7806)				
Yes	1070 (13.7%)	379 (35.5%)	688 (64.5%)	<0.001 ²
No	6736 (86.3%)	4788 (71.2%)	1941 (28.8%)	
Frequency of Feeling Worried/Nervous/Anxious (N = 7785)				
Infrequent	4656 (59.8%)	3274 (70.4%)	1378 (29.6%)	<0.001 ²
Frequent	3129 (40.2%)	1884 (60.2%)	1245 (39.8%)	

²p-value for chi-square test or Fishers exact test; ³p-value for t-test

Numbers may not sum to the total sample size because of missing data. Percentages are row percentages. PHQ-9 scores were missing for 33.3% of participants; therefore, the denominator for depression-related variables is n = 5,210. Denominators for other variables vary as indicated.

Acronyms: Med (median); IQR (interquartile range); PHQ-9 = Patient Health Questionnaire-9; NHANES = National Health and Nutrition Examination Survey

Table 2 presents the bivariable associations between difficulty walking or climbing steps and key sociodemographic characteristics, the main outcome, and covariates among 7,809 U.S. adults aged 20 years and older. Overall, 33.7% of participants reported any difficulty walking or climbing steps. Participants with any difficulty walking had a higher mean PHQ-9 continuous depression score compared to those with no difficulty (5.5 vs. 3.3, $p < 0.001$), and a higher prevalence of binary depression (52.8% vs. 47.2%, $p < 0.001$). The prevalence of difficulty walking was higher among females than males (37.6% vs. 29.0%, $p < 0.001$), among older participants (mean age 62.5 vs. 49.1 years, $p < 0.001$), and across racial/ethnic groups ($p < 0.001$). Those with less than a high school education had a higher prevalence compared to those with some college or above (47.1% vs. 29.2%, $p < 0.001$), and those who were widowed, divorced, or separated had a higher prevalence compared to those who were married (50.3% vs. 29.4%, $p < 0.001$). Participants born in the US had a higher prevalence than those born outside the US (35.3% vs. 27.7%, $p < 0.001$), and those at or below the poverty level had a higher prevalence compared to those above the poverty level (45.1% vs. 31.5%, $p < 0.001$). Among covariates, difficulty walking was more prevalent among those with diabetes (64.5% vs. 28.8%, $p < 0.001$), those with high cholesterol (45.4% vs. 25.9%, $p < 0.001$), and those who frequently felt worried, nervous, or anxious (39.8% vs. 29.6%, $p < 0.001$).

Table 3a. Bivariable associations between depression (PHQ-9 binary score) and difficulty walking or climbing steps, key sociodemographic characteristics, and other covariates, NHANES 2021-2023 (n = 7,809)

		Depression (PHQ-9) sum score dichotomized		
Sociodemographic Characteristics	Total N (%)	No N (%)	Yes N (%)	p-value
Gender (N = 7809)				
Male	3488 (44.7%)	2121 (89.7%)	243 (10.3%)	<0.001 ²
Female	4321 (55.3%)	2428 (85.3%)	418 (14.7%)	
Age (years) (N = 7809)				
Mean (SD, range)	53.6 (17.5, 20 - 80)	54.5 (16.9, 20 - 80)	48.7 (17.4, 20 - 80)	<0.001 ³
Median age (IQR)	57 (38 - 68)	58 (40 - 68)	50 (32 - 63)	
Race/Ethnicity (N = 7809)				
Mexican American	545 (7.0%)	284 (86.1%)	46 (13.9%)	0.002 ²
Other Hispanic	780 (10.0%)	433 (86.8%)	66 (13.2%)	
Non-Hispanic White	4555 (58.3%)	2830 (88.0%)	385 (12.0%)	
Non-Hispanic Black	995 (12.7%)	514 (85.9%)	84 (14.1%)	
Non-Hispanic Asian	422 (5.4%)	211 (92.5%)	17 (7.5%)	
Other/Multi-Racial	512 (6.6%)	277 (81.5%)	63 (18.5%)	
Education level (N = 7783)				
Less than High School	1039 (13.3%)	452 (82.5%)	96 (17.5%)	<0.001 ²
High School Graduate/GED	1749 (22.5%)	906 (84.6%)	165 (15.4%)	
Some College or College Graduate	4995 (64.2%)	3190 (88.9%)	399 (11.1%)	
Marital Status (N = 7783)				
Married/Living with partner	4136 (53.1%)	2567 (91.1%)	250 (8.9%)	<0.001 ²
Widowed/Divorced/Separated	2022 (26.0%)	1106 (84.9%)	197 (15.1%)	
Never Married	1625 (20.9%)	873 (80.4%)	213 (19.6%)	
Country of Birth (N = 7793)				
Born in 50 US states or Washington DC	6219 (79.8%)	3703 (86.7%)	570 (13.3%)	0.003 ²
Others	1574 (20.2%)	846 (90.3%)	91 (9.7%)	
Ratio of family income to poverty (N = 6489)				
<=1				<0.001 ²
>1	1044 (16.1%) 5445 (83.9%)	510 (77.3%) 3482 (88.7%)	150 (22.7%) 444 (11.3%)	
Difficulty Walking or Climbing Steps (N = 7797)				
No Difficulty	5167 (66.3%)	3175 (91.1%)	312 (8.9%)	<0.001 ²
Any Difficulty	2630 (33.7%)	1372 (79.7%)	349 (20.3%)	
High Cholesterol (N = 7755)				
Yes	3052 (39.4%)	1934 (87.9%)	265 (12.1%)	<0.277 ²
No	4703 (60.6%)	2595 (86.9%)	390 (13.1%)	
Diabetes Condition (N = 7806)				
Yes	1070 (13.7%)	579 (80.9%)	137 (19.1%)	<0.001 ²
No	6736 (86.3%)	3970 (88.3%)	524 (11.7%)	

		Depression (PHQ-9) sum score dichotomized		
Sociodemographic Characteristics	Total N (%)	No N (%)	Yes N (%)	<i>p</i> -value
Frequency of Feeling Worried/Nervous/Anxious (N = 7785)				
Infrequent	4656 (59.8%)	2945 (95.9%)	127 (4.1%)	<0.001 ²
Frequent	3129 (40.2%)	1599 (75.0%)	534 (25.0%)	

²*p*-value for chi-square test or Fishers exact test; ³*p*-value for t-test

Numbers may not sum to the total sample size because of missing data. Percentages are row percentages. PHQ-9 scores were missing for 33.3% of participants; therefore, the denominator for depression-related variables is n = 5,210. Denominators for other variables vary as indicated.

Acronyms: Med (median); IQR (interquartile range); PHQ-9 = Patient Health Questionnaire-9; NHANES = National Health and Nutrition Examination Survey

Table 3a presents the bivariable associations between binary depression status and key sociodemographic characteristics, the main exposure, and covariates among 7,809 U.S. adults aged 20 years and older. Overall, 12.7% of participants met the criteria for depression. Participants with any difficulty walking had a significantly higher prevalence of depression compared to those with no difficulty (20.3% vs. 8.9%, $p < 0.001$). The prevalence of depression was higher among females than males (14.7% vs. 10.3%, $p < 0.001$) and among younger participants (mean age 48.7 vs. 54.5 years, $p < 0.001$). Depression prevalence also differed across racial/ethnic groups ($p = 0.002$). Those with less than a high school education had a higher prevalence than those with some college or above (17.5% vs. 11.1%, $p < 0.001$), and those who were never married had a higher prevalence compared to those who were married (19.6% vs. 8.9%, $p < 0.001$). Participants born in the US had a higher prevalence than those born outside the US (13.3% vs. 9.7%, $p = 0.003$), and those at or below the poverty level had a higher prevalence compared to those above the poverty level (22.7% vs. 11.3%, $p < 0.001$). Among covariates, diabetes was significantly associated with a higher prevalence of depression (19.1% vs. 11.7%, $p < 0.001$), while high cholesterol was not (12.1% vs. 13.1%, $p = 0.277$). Those who frequently felt worried, nervous, or anxious had a substantially higher prevalence of depression compared to those who infrequently felt anxious (25.0% vs. 4.1%, $p < 0.001$), suggesting that anxiety frequency may modify the association between walking difficulty and depression. This should be examined further in stratified analyses.

Taken together with Table 2, these findings support diabetes as a potential confounder given its significant association with both the exposure ($p < 0.001$) and the binary outcome ($p < 0.001$). High cholesterol does not meet the criteria for confounding because it was not significantly associated with the binary depression outcome ($p = 0.277$), despite its association with the exposure ($p < 0.001$). Therefore, high cholesterol should be considered for removal from the final multivariable model. Overall, the bivariable findings support diabetes as the only pre-specified confounder that should be adjusted for in the multivariable analysis. Other factors not considered in the research question that may warrant consideration as potential confounders include number of prescription medicines taken in the past 30 days, type of work done last week, and chronic pain conditions.

Table 3b. Bivariable associations between depression (PHQ-9 continuous score) and difficulty walking or climbing steps, key sociodemographic characteristics, and other covariates, NHANES 2021-2023 (n = 7,809)

		Depression (PHQ-9) Sum score	
Sociodemographic Characteristics	Total N (%)	Mean (SD)	p-value
Gender (N = 7809)			
Male	3488 (44.7%)	3.5 (4.5)	<0.001 ³
Female	4321 (55.3%)	4.5 (4.8)	
Age (years) (N = 7809)			
Mean (SD, range)	53.6 (17.5, 20 - 80)	4.0 (4.7)	<0.001 ⁵
Median age (IQR)	57 (38 – 68)	Pearson's r = -0.1619 Spearman's rho = -0.1776	<0.001 ⁶
Race/Ethnicity (N = 7809)			
Mexican American	545 (7.0%)	4.1 (4.9)	0.0001 ⁴
Other Hispanic	780 (10.0%)	4.4 (4.9)	
Non-Hispanic White	4555 (58.3%)	3.9 (4.6)	
Non-Hispanic Black	995 (12.7%)	4.1 (4.8)	
Non-Hispanic Asian	422 (5.4%)	3.1 (3.9)	
Other/Multi-Racial	512 (6.6%)	4.9 (5.3)	
Education level (N = 7783)			
Less than High School	1039 (13.3%)	4.9 (5.6)	<0.001 ⁴
High School Graduate/GED	1749 (22.5%)	4.4 (5.0)	
Some College or College Graduate	4995 (64.2%)	3.8 (4.4)	
Marital Status (N = 7783)			
Married/Living with partner	4136 (53.1%)	3.3 (4.0)	<0.001 ⁴
Widowed/Divorced/Separated	2022 (26.0%)	4.6 (5.1)	
Never Married	1625 (20.9%)	5.3 (5.3)	
Country of Birth (N = 7793)			
Born in 50 US states or Washington DC	6219 (79.8%)	4.2 (4.7)	<0.001 ³
Others	1574 (20.2%)	3.5 (4.4)	
Ratio of family income to poverty (N = 6489)			
<=1			<0.001 ³
>1	1044 (16.1%)	5.7 (5.7)	
	5445 (83.9%)	3.8 (4.5)	
Difficulty Walking or Climbing Steps (N = 7797)			
No Difficulty			<0.001 ³
Any Difficulty	5167 (66.3%) 2630 (33.7%)	3.3 (4.2) 5.5 (5.3)	
High Cholesterol (N = 7755)			
Yes	3052 (39.4%)	4.0 (4.7)	0.9094 ³
No	4703 (60.6%)	4.0 (4.7)	
Diabetes Condition (N = 7806)			
Yes	1070 (13.7%)	5.0 (5.4)	<0.001 ³
No	6736 (86.3%)	3.9 (4.5)	

		Depression (PHQ-9) Sum score	
Sociodemographic Characteristics	Total N (%)	Mean (SD)	<i>p</i> -value
Frequency of Feeling Worried/Nervous/Anxious (N = 7785)			
Infrequent	4656 (59.8%)	2.3 (3.0)	<0.001 ³
Frequent	3129 (40.2%)	6.5 (5.5)	

³*p*-value for t-test; ⁴*p*-value for ANOVA; ⁵*p*-value from Pearson's correlation; ⁶*p*-value from Spearman's correlation

Numbers may not sum to the total sample size because of missing data. PHQ-9 scores were missing for 33.3% of participants; therefore, the denominator for depression-related variables is n = 5,210. Denominators for other variables vary as indicated.

Acronyms: Med (median); IQR (interquartile range); PHQ-9 = Patient Health Questionnaire-9; NHANES = National Health and Nutrition Examination Survey

Table 3b presents the bivariable associations between the continuous PHQ-9 depression score and key sociodemographic characteristics, the main exposure, and covariates among 7,809 U.S. adults aged 20 years and older. The overall mean PHQ-9 score was 4.0 (SD = 4.7). The PHQ-9 score distribution was right-skewed, with several outlier values observed beyond the upper whisker of the boxplot (scores 15-27), representing a small proportion of participants with severe depression symptoms. These values were kept because they fall within the valid PHQ-9 range of 0-27. Participants with any difficulty walking had a significantly higher mean PHQ-9 score compared to those with no difficulty (5.5 vs. 3.3, $p < 0.001$), consistent with the findings in Table 3a. Females had a higher mean PHQ-9 score than males (4.5 vs. 3.5, $p < 0.001$). Age was significantly and inversely associated with PHQ-9 score based on both Pearson's correlation ($p < 0.001$) and Spearman's correlation ($p < 0.001$). Mean PHQ-9 scores differed significantly across racial/ethnic groups ($p = 0.0001$), were higher among those with less than a high school education compared to those with some college or above (4.9 vs. 3.8, $p < 0.001$), higher among those who were never married compared to those who were married (5.3 vs. 3.3, $p < 0.001$), and higher among those at or below the poverty level compared to those above the poverty level (5.7 vs. 3.8, $p < 0.001$).

Among covariates, diabetes was significantly associated with a higher mean PHQ-9 score (5.0 vs. 3.9, $p < 0.001$), while high cholesterol was not (4.0 vs. 4.0, $p = 0.909$). Those who frequently felt worried, nervous, or anxious had a substantially higher mean PHQ-9 score compared to those who infrequently felt anxious (6.5 vs. 2.3, $p < 0.001$), further suggesting that anxiety frequency may modify the association between walking difficulty and depression severity. This should be examined further in stratified analyses. Consistent with Table 2 and Table 3a, these findings support diabetes as a potential confounder that should be kept in the multivariable model. High cholesterol does not meet the criteria for confounding given its non-significant association with the continuous PHQ-9 score ($p = 0.909$), and should be considered for removal from the final multivariable model. Overall, the bivariable findings support diabetes as the only pre-specified confounder that should be adjusted for in the multivariable analysis. Sociodemographic factors including age, gender, marital status, education, race/ethnicity, country of birth, and income to poverty ratio all showed significant associations with the continuous depression outcome and with the exposure (Table 2), and should be adjusted for in the multivariable regression analyses. Other factors not considered in the research question that may warrant consideration as potential confounders include number of prescription medicines taken in the past 30 days, type of work done last week, and chronic pain conditions.

Multivariable Analysis

Table 4a. Multivariable logistic regression model examining difficulty walking or climbing steps and depression among U.S. adults aged 20 years and older, NHANES 2021-2023

	Crude OR (95% CI)	Full model 1 Adjusted OR (95% CI)	Full model 2 Adjusted OR (95% CI)
Gender (ref: Male)			
Male	Referent (1.00)	Referent (1.00)	Referent (1.00)
Female	1.5 (1.2, 1.8)	1.2 (1.0, 1.5)	1.2 (1.0, 1.5)
Age Group, years (ref: 20–39)			
20-39	Referent (1.00)	Referent (1.00)	Referent (1.00)
40-59	0.6 (0.5, 0.8)	0.6 (0.5, 0.8)	0.6 (0.5, 0.8)
60-74	0.6 (0.4, 0.7)	0.4 (0.3, 0.6)	0.4 (0.3, 0.6)
75+	0.4 (0.3, 0.5)	0.3 (0.2, 0.5)	0.3 (0.2, 0.5)
Race/Ethnicity (ref: Non-Hispanic White)			
Non-Hispanic White	Referent (1.00)	Referent (1.00)	Referent (1.00)
Mexican American	1.2 (0.8, 1.7)	1.2 (0.8, 1.8)	1.2 (0.8, 1.8)
Other Hispanic	1.2 (0.9, 1.6)	1.1 (0.7, 1.6)	1.1 (0.7, 1.6)
Non-Hispanic Black	1.2 (0.9, 1.6)	1.0 (0.7, 1.4)	1.0 (0.7, 1.4)
Non-Hispanic Asian	0.5 (0.3, 0.9)	0.9 (0.5, 1.7)	0.9 (0.5, 1.8)
Other/Multiracial	1.6 (1.2, 2.2)	1.2 (0.8, 1.7)	1.2 (0.8, 1.7)
Marital Status (ref: Married/Living with partner)			
Married/Living with partner	Referent (1.00)	Referent (1.00)	Referent (1.00)
Widowed/Divorced/Separated	1.9 (1.5, 2.3)	1.5 (1.2, 1.9)	1.5 (1.2, 1.9)
Never married	2.5 (2.0, 3.1)	1.6 (1.3, 2.1)	1.6 (1.3, 2.1)
Education (ref: Less than high school)			
Less than high school	Referent (1.00)	Referent (1.00)	Referent (1.00)
High school/GED	0.9 (0.7, 1.2)	1.0 (0.7, 1.4)	1.0 (0.7, 1.4)
Some college or above	0.6 (0.5, 0.8)	0.7 (0.5, 1.0)	0.7 (0.5, 1.0)
Country of Birth (ref: Born in US)			
Born in US	Referent (1.00)	Referent (1.00)	Referent (1.00)
Born outside US	0.6 (0.5, 0.8)	0.8 (0.6, 1.1)	0.8 (0.6, 1.1)
Income-to-Poverty Ratio (ref: Above poverty, >1)			
Above poverty (>1)	Referent (1.00)	Referent (1.00)	Referent (1.00)
At or below poverty (≤1)	2.4 (1.9, 2.9)	1.3 (1.0, 1.7)	1.3 (1.0, 1.7)
Difficulty Walking or Climbing Steps (ref: No difficulty)			
No difficulty	Referent (1.00)	Referent (1.00)	Referent (1.00)
Some difficulty	1.9 (1.5, 2.3)	2.1 (1.6, 2.7)	2.6 (1.6, 4.2)
A lot of difficulty	4.7 (3.7, 6.0)	5.0 (3.7, 6.9)	6.3 (3.7, 10.9)
Cannot do at all	5.7 (2.9, 12.3)	6.2 (2.7, 14.3)	14.7 (4.7, 45.6)
High Cholesterol (ref: No)			
No	Referent (1.00)	Referent (1.00)	Referent (1.00)
Yes	0.9 (0.7, 1.1)	0.9 (0.7, 1.2)	0.9 (0.7, 1.2)
Diabetes (ref: No/Borderline)			
No/Borderline	Referent (1.00)	Referent (1.00)	Referent (1.00)

Yes	1.8 (1.4, 2.2)	1.6 (1.2, 2.2)	1.6 (1.2, 2.2)
Anxiety Frequency (ref: Less frequent/Never)			
Less frequent/Never	Referent (1.00)	Referent (1.00)	Referent (1.00)
Daily/Weekly	8.0 (6.4, 9.9)	6.6 (5.3, 8.3)	7.8 (5.6, 10.9)
Walking Difficulty × Anxiety Frequency			
No difficulty × Less frequent/Never	NA	NA	Referent (1.00)
Some difficulty × Daily/Weekly	NA	NA	0.8 (0.4, 1.3)
A lot of difficulty × Daily/Weekly	NA	NA	0.7 (0.4, 1.4)
Cannot do at all × Daily/Weekly	NA	NA	0.2 (0.1, 1.1)

OR = Odds ratio; CI = Confidence interval.

Table 4a presents the crude and adjusted logistic regression models examining the association between difficulty walking or climbing steps and binary depression among U.S. adults aged 20 years and older. In the crude model, difficulty walking or climbing steps was strongly associated with depression. Compared with participants who reported no difficulty walking or climbing steps, participants with some difficulty had higher odds of depression, participants with a lot of difficulty had even higher odds, and participants who could not do this activity at all had the highest odds of depression. This shows a graded pattern, where greater walking difficulty was associated with higher odds of depression.

After adjustment for gender, age group, race/ethnicity, marital status, education, country of birth, income-to-poverty ratio, high cholesterol, diabetes, and anxiety frequency, the association remained strong. In Full model 1, participants with some difficulty walking or climbing steps had 2.1 times the odds of depression compared with those with no difficulty. Participants with a lot of difficulty had 5.0 times the odds of depression, and those who could not do this activity at all had 6.2 times the odds of depression. Frequent anxiety symptoms were also strongly associated with depression. Participants who reported feeling worried, nervous, or anxious daily or weekly had 6.6 times the odds of depression compared with those who reported these symptoms less frequently or never. In Full model 2, the interaction terms between walking difficulty and anxiety frequency were added. The main association between walking difficulty and depression remained elevated, but the interaction terms had confidence intervals that included 1.0. These findings suggest that walking difficulty and frequent anxiety symptoms were both strongly associated with depression, but there was limited evidence that anxiety frequency modified the association between walking difficulty and depression.

Table 4b. Multivariable linear regression model examining difficulty walking or climbing steps and PHQ-9 depression score among U.S. adults aged 20 years and older, NHANES 2021-2023

	Unadjusted model		Full model 1		Full model 2	
	beta (SE)	p-value	beta (SE)	p-value	beta (SE)	p-value
Gender (ref: Male)						
Male	Ref		Ref		Ref	
Female	1.0 (0.1)	<0.001	0.4 (0.1)	<0.001	0.4 (0.1)	<0.001
Age Group, years (ref: 20–39)						
20-39	Ref		Ref		Ref	
40-59	-1.3 (0.2)	<0.001	-1.0 (0.2)	<0.001	-1.1 (0.2)	<0.001
60-74	-1.7 (0.2)	<0.001	-1.8 (0.2)	<0.001	-1.9 (0.2)	<0.001
75+	-2.2 (0.2)	<0.001	-2.1 (0.2)	<0.001	-2.2 (0.2)	<0.001
Race/Ethnicity (ref: Non-Hispanic White)						
Non-Hispanic White	Ref		Ref		Ref	
Mexican American	0.2 (0.3)	0.475	0.2 (0.3)	0.567	0.2 (0.3)	0.530
Other Hispanic	0.6 (0.2)	0.012	0.4 (0.2)	0.095	0.4 (0.2)	0.119
Non-Hispanic Black	0.2 (0.2)	0.389	-0.1 (0.2)	0.652	-0.1 (0.2)	0.594
Non-Hispanic Asian	-1.1 (0.3)	0.002	0.0 (0.3)	0.917	0.0 (0.3)	0.943
Other/Multiracial	1.1 (0.3)	<0.001	0.4 (0.2)	0.129	0.4 (0.2)	0.137
Marital Status (ref: Married/Living with partner)						
Married/Living with partner	Ref		Ref		Ref	
Widowed/Divorced/Separated	1.4 (0.2)	<0.001	0.8 (0.1)	<0.001	0.8 (0.1)	<0.001
Never married	2.1 (0.2)	<0.001	1.0 (0.2)	<0.001	1.0 (0.2)	<0.001
Education (ref: Less than high school)						
Less than high school	Ref		Ref		Ref	
High school/GED	-0.6 (0.3)	0.040	-0.3 (0.2)	0.260	-0.3 (0.2)	0.205
Some college or above	-1.2 (0.2)	<0.001	-0.6 (0.2)	0.004	-0.6 (0.2)	0.003
Country of Birth (ref: Born in US)						
Born in US	Ref		Ref		Ref	
Born outside US	-0.8 (0.2)	<0.001	-0.3 (0.2)	0.151	-0.3 (0.2)	0.128
Income-to-Poverty Ratio (ref: Above poverty, >1)						
Above poverty (>1)	Ref		Ref		Ref	
At or below poverty (≤ 1)	1.9 (0.2)	<0.001	0.6 (0.2)	0.001	0.6 (0.2)	0.002
Difficulty Walking or Climbing Steps (ref: No difficulty)						
No difficulty	Ref		Ref		Ref	
Some difficulty	1.6 (0.2)	<0.001	1.5 (0.2)	<0.001	1.3 (0.2)	<0.001
A lot of difficulty	3.8 (0.2)	<0.001	3.3 (0.2)	<0.001	2.1 (0.3)	<0.001
Cannot do at all	4.8 (0.7)	<0.001	3.9 (0.7)	<0.001	4.1 (1.0)	<0.001
High Cholesterol (ref: No)						
No	Ref		Ref		Ref	
Yes	0.0 (0.1)	0.752	0.3 (0.1)	0.042	0.3 (0.1)	0.043
Diabetes (ref: No/Borderline)						
No/Borderline	Ref		Ref		Ref	
Yes	1.1 (0.2)	<0.001	0.7 (0.2)	<0.001	0.7 (0.2)	<0.001

Anxiety Frequency (ref: Less frequent/Never)						
Less frequent/Never	Ref		Ref		Ref	
Daily/Weekly	4.2 (0.1)	<0.001	3.6 (0.1)	<0.001	3.2 (0.2)	<0.001
Walking Difficulty × Anxiety Frequency						
No difficulty × Less frequent/Never	NA	NA	NA	NA	Ref	
Some difficulty × Daily/Weekly	NA	NA	NA	NA	0.6 (0.3)	0.033
A lot of difficulty × Daily/Weekly	NA	NA	NA	NA	2.3 (0.4)	<0.001
Cannot do at all × Daily/Weekly	NA	NA	NA	NA	-0.3 (1.4)	0.857

SE = Standard error. Ref = Referent category.

Values that are presented as 0.0 in the table are because all the numbers are kept to one decimal point, the true values are in the range of 0.01-0.04.

Table 4b presents the crude and adjusted linear regression models examining the association between difficulty walking or climbing steps and continuous PHQ-9 depression score among U.S. adults aged 20 years and older. In the unadjusted model, greater difficulty walking or climbing steps was associated with higher mean PHQ-9 depression scores. Compared with participants who reported no difficulty walking or climbing steps, participants with some difficulty had higher average PHQ-9 scores, while participants with a lot of difficulty or who could not do this activity at all had even higher average scores. This pattern is consistent with the bivariable analysis and suggests that depression severity increased as walking difficulty became more severe.

After adjustment for sociodemographic characteristics and health-related covariates in Full model 1, walking difficulty remained associated with higher PHQ-9 depression scores. Participants with some difficulty, a lot of difficulty, and those who could not walk or climb steps at all all had higher adjusted mean PHQ-9 scores compared with participants with no difficulty. Frequent anxiety symptoms were also strongly associated with PHQ-9 score. Participants who reported feeling worried, nervous, or anxious daily or weekly had higher adjusted mean PHQ-9 scores than those who reported these symptoms less frequently or never. In Full model 2, the interaction between walking difficulty and anxiety frequency was added. The interaction terms did not show strong evidence that the association between walking difficulty and PHQ-9 score differed by anxiety frequency. Overall, the linear regression results support the logistic regression results: greater difficulty walking or climbing steps was associated with higher depression severity, and frequent anxiety symptoms were also strongly associated with depression, but there was limited evidence for effect modification by anxiety frequency.

Discussion

This study found that difficulty walking or climbing steps was consistently associated with higher depression severity among U.S. adults aged 20 years and older. The association was present in bivariable analyses, persisted after multivariable adjustment, and showed a graded pattern across increasing severity of walking difficulty. These findings are consistent with previous studies showing that physical functioning limitations, disability domains, and walking impairment are related to worse mental health and higher depressive symptom burden (Becofsky et al., 2013; Meshkat et al., 2025; Rezvani et al., 2022; Thompson et al., 2012; Zhang & Jiang, 2023).

The findings also support the importance of anxiety frequency in the relationship between walking difficulty and depression. Frequent feelings of being worried, nervous, or anxious were strongly associated with both binary depression and continuous PHQ-9 score. In the multivariable linear regression model, the interaction terms suggested potential effect modification on the additive scale, while the logistic regression interaction estimates were imprecise and did not provide clear evidence of multiplicative effect modification. Taken together, these findings are consistent with prior evidence that anxiety and depression are closely linked to disability, but the exact scale and magnitude of effect modification require additional study (Brenes et al., 2008).

Several strengths support this analysis. NHANES uses a large, probability-based sample designed to represent the U.S. civilian, non-institutionalized population. Data collection includes standardized questionnaires and examination components, and depression was measured using the validated PHQ-9 instrument (Centers for Disease Control and Prevention, 2026; Kroenke et al., 2001). The analysis also evaluated both binary depression and continuous depression severity, which provides a more complete view of the association than either outcome alone.

This study also has important limitations. The cross-sectional design prevents establishing temporality, so it is not possible to determine whether walking difficulty preceded depression, depression contributed to reduced mobility, or both processes occurred together. Missing PHQ-9 data were substantial, with 33.3% missing among adults aged 20 years and older. If participants with missing PHQ-9 scores differed systematically in both walking difficulty and depression, selection bias may have been introduced. Self-reported walking difficulty, anxiety frequency, diabetes, high cholesterol, and smoking-related variables may also introduce information bias due to recall error or reporting differences. In addition, the analysis

did not fully account for NHANES complex survey weights in the presented tables, which may limit national generalizability of the estimates.

These biases may affect both internal and external validity. Selection bias from missing PHQ-9 data could bias the measures of association if missingness was related to both mobility limitation and depressive symptoms. Information bias could lead to exposure or covariate misclassification, which may either attenuate or exaggerate associations depending on whether misclassification differed by depression status. External validity is also limited because NHANES represents the civilian, non-institutionalized population; findings may not generalize to institutionalized adults, including nursing home residents, who may have more severe mobility limitations and depression burden.

Several strategies were used to reduce the impact of these biases. Refused and do not know responses were treated as missing rather than being analyzed as meaningful categories. Valid PHQ-9 scores were calculated only from complete item responses, and implausible values were checked against valid score ranges before being retained. Multivariable regression models adjusted for key sociodemographic and health-related variables to reduce confounding. However, because of the cross-sectional design, self-reported measures, missing data, and possible unmeasured confounding, the findings should be interpreted as evidence of association rather than causation.

Future research should use longitudinal data to examine whether walking difficulty predicts later depression and whether depression predicts worsening physical functioning. Additional studies should also evaluate whether interventions that improve mobility, reduce anxiety symptoms, or address both simultaneously can reduce depression severity among adults with physical functioning limitations.

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